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Mind Scan Analyzer: Classifying Psychiatric Disorder Groups from Resting-State EEG Features with a Browser-Deployed Random Forest

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ABSTRACT. Mind Scan Analyzer is a web tool for exploring machine-learning classification of psychiatric disorder groups from resting-state quantitative EEG (qEEG) features. This note documents the data, preprocessing, feature selection, model, deployment design and results of the project. A Random Forest classifier using ten ANOVA-selected features is trained on the public BRMH dataset (945 subjects, seven diagnostic groups) and exported to JSON so that prediction runs entirely in the browser. On a stratified 20% test split (189 subjects) it reaches 98.41% accuracy with a macro F1 of 0.98; per-class scores, limitations and the next evaluation steps are discussed.

1 Objective

To build an end-to-end, reproducible pipeline — from tabular EEG features to an interactive, server-free prediction interface — that can be used to teach applied machine learning on health data and to explore EEG-based screening research. The tool is not a diagnostic device.

2 Data

The data come from Park et al. [1], who collected resting-state EEG, IQ scores and medical records of 945 subjects — 850 patients with major psychiatric disorders and 95 healthy controls. A public copy (`EEG.machinelearing_data_BRMH.csv`) is available on Kaggle [2].

- **Columns (1,149):** identifiers (`no.`, `eeg.date`), demographics (sex, age, education, IQ) and qEEG features — absolute band power (`AB.*`) and inter-channel coherence (`COH.*`) across frequency bands.
- **Target:** `main.disorder` (seven groups). The finer `specific.disorder` label is excluded.

DIAGNOSTIC GROUP	N	%
Mood disorder	266	28.1
Addictive disorder	186	19.7
Trauma- and stress-related	128	13.5
Schizophrenia	117	12.4
Anxiety disorder	107	11.3
Healthy control	95	10.1
Obsessive-compulsive disorder	46	4.9
Total	**945**	**100**

Table 1. Class distribution of `main.disorder`. The largest group is 5.8× the smallest, so macro-averaged metrics are essential.

3 Pipeline

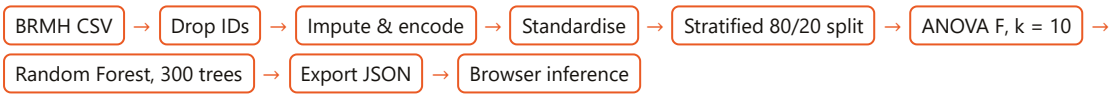


Figure 1. Training and deployment pipeline as implemented in `scripts/export_model.py`.

1. **Cleaning.** Identifier columns and the secondary label are dropped, as are columns that are entirely empty.
2. **Imputation and encoding.** Numeric features: median imputation. Categorical features (e.g. sex): most-frequent imputation followed by label encoding. Target labels are encoded to seven integer classes.
3. **Scaling.** All features are standardised to zero mean and unit variance (z-scores).
4. **Split.** A stratified 80/20 split with a fixed seed (`random_state = 42`) gives 756 training and 189 test subjects with the class proportions of Table 1.
5. **Feature selection.** `SelectKBest` with the ANOVA F-statistic keeps the ten most discriminative features. It is fitted on the training split only, so test labels never inform the selection.
6. **Model.** A Random Forest [3] with 300 trees and `class_weight = "balanced"` (to counter the imbalance in Table 1) is trained with scikit-learn [4]. Random Forests capture non-linear interactions with little tuning and expose feature importances, which suits an exploratory, teaching-oriented tool.

4 Deployment: inference in the browser

The fitted pipeline is serialised to JSON: the scaler's means and scales, the indices of the selected features and, for every tree, its split features, thresholds, child pointers and per-class leaf counts. A TypeScript module (`src/model/inference.ts`) re-implements scikit-learn's `predict_proba`: it applies the same preprocessing to an uploaded CSV, walks each tree, normalises the leaf counts and averages them across trees. The React + Vite interface offers Upload & Predict, Model Performance, Model Insights (feature importance) and Dataset views. Because no server is involved, uploaded data never leaves the user's device.

5 Results

On the held-out test split (189 subjects, `random_state = 42`) the model classifies 186 subjects correctly: **accuracy 98.41%**, **macro F1 0.98** and weighted F1 0.98. The three errors are two anxiety-disorder subjects and one healthy control assigned to other groups (Table 2). The results were presented and demonstrated during the MCA final-year evaluation (viva).

For context, Park et al. [1] report *binary* patient-versus-control accuracies of about 74.5% (obsessive-compulsive disorder) to 93.8% (schizophrenia) with a different protocol, so the figures are not directly comparable. Because they come from a single split, they should be confirmed with cross-validation and, ideally, an external dataset (Section 6).

CLASS	PREC.	REC.	F1	N
Addictive disorder	1.00	1.00	1.00	37
Anxiety disorder	1.00	0.91	0.95	22
Healthy control	1.00	0.95	0.97	19
Mood disorder	0.98	1.00	0.99	53
Obsessive-compulsive	1.00	1.00	1.00	9
Schizophrenia	0.96	1.00	0.98	23
Trauma- and stress-related	0.96	1.00	0.98	26
Macro avg · accuracy 98.41%	**0.99**	**0.98**	**0.98**	**189**

Table 2. Per-class precision, recall and F1 on the stratified test split.

6 Limitations and planned improvements

- **Preprocessing leakage.** The imputer and scaler are fitted on all 945 rows before the split. Wrapping every step in a scikit-learn `Pipeline` fitted on training folds only removes this mild leakage.
- **Single split.** One 80/20 split gives a high-variance estimate, and the OCD class has only nine test subjects. Next step: stratified 5-fold cross-validation (mean ± SD of macro-F1 and balanced accuracy) with nested tuning of k, plus baselines (majority class, logistic regression).
- **Imbalance.** Obsessive-compulsive disorder has only 46 subjects; macro metrics and, if needed, resampling should accompany class weighting.
- **Generalisation.** All data come from one dataset; performance on other EEG systems, clinics and populations is unknown.
- **Ethics.** Outputs are not diagnoses. The tool is intended for research and teaching only and must not inform clinical decisions.

7 Reproducibility

Code: github.com/its-aditisingh/Mind-Scan-Analyzer (MIT licence). Training: `python scripts/export_model.py EEG.machinelearing_data_BRMH.csv`, which writes `src/model/model.json`; `npm install && npm run dev` starts the interface.

8 References

1. Park, S. M., et al. (2021). Identification of major psychiatric disorders from resting-state electroencephalography using a machine learning approach. *Frontiers in Psychiatry*, 12, 707581. https://doi.org/10.3389/fpsyt.2021.707581
2. EEG Psychiatric Disorders Dataset (public copy of the data in [1]). Kaggle. https://www.kaggle.com/datasets/shashwatwork/eeg-psychiatric-disorders-dataset
3. Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
4. Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.